



AI / LLM SYSTEMS ENGINEER - INNOVATION INTELLIGENCE

(Incubator AIC - GBU / AI Research & Engineering Intern)

ABOUT OPEN IDEA

Open Idea is an AI-powered innovation platform that turns ideas into real apps and prototypes—no coding required.

The Problem: You have an idea, but execution is broken. Knowledge is scattered across 30,000+ journals and 420 million+ code repositories. Finding relevant research is impossible. Hiring developers is expensive. Building from scratch wastes time and money. Result? 90% of Indian startups fail within 3 years, with ₹2,000 Crore wasted annually.

How Open Idea Works: Start with your idea. Our platform offers three integrated components to turn it into reality:

- 1. Research:** Access millions of research papers, datasets, and open-source code in one unified platform. Our AI helps you discover existing solutions instead of starting from zero.
- 2. Build:** Use our AI-powered no-code workspace to remix proven components and create production-ready prototypes. No coding skills required. No expensive developers needed.
- 3. Collaborate:** Connect with a global community of experts, researchers, and innovators who can help refine and accelerate your project.

The Result: What traditionally takes months now takes minutes. From idea to working prototype—fast.

From Idea to App: Search your idea → AI finds relevant research, code, and datasets → Remix components in our no-code workspace → Collaborate with experts → Launch your prototype. That's innovation, democratized.

ABOUT THE ROLE

This is a highly technical role at the core of Open Idea's AI infrastructure. You'll be responsible for the intelligence layer that powers our platform—designing AI systems that turn user ideas into reliable, executable prototypes. We're looking for an AI/ML engineer who understands both the theoretical foundations and practical applications of LLMs, prompt engineering, and semantic search. You'll optimize AI performance, ensure output quality, and build the systems that make innovation accessible to everyone.

KEY RESPONSIBILITIES

- **Prompt Engineering & Pipeline Design (Q1 2026):** Build robust prompt pipelines that convert user ideas into structured, buildable project specifications with 90%+ success rate.
- **Semantic Search & Knowledge Graph:** Develop and optimize semantic search algorithms to help users discover relevant research papers, datasets, and code from our unified knowledge base.
- **AI Output Validation & Safety:** Implement validation layers to ensure AI-generated code, recommendations, and configurations are executable, secure, and aligned with user intent.
- **Model Selection & Fine-Tuning:** Evaluate, compare, and select the best LLMs (GPT-4, Claude, Gemini, open-source models) for different use cases. Fine-tune models when necessary.
- **Performance Optimization:** Reduce AI response latency to ≤ 1.5 seconds while maintaining quality. Optimize token usage and API costs for scalability.
- **AI Monitoring & Debugging:** Track AI system performance, identify failure patterns, debug prompt issues, and implement failover mechanisms to ensure reliability.
- **Research & Innovation:** Stay updated on latest AI/LLM developments and experiment with new techniques (RAG, function calling, agentic workflows, multimodal AI) to enhance platform capabilities.

EXPECTED DELIVERABLES

- **Prompt Library:** Production-ready prompt templates for idea generation, code synthesis, project structuring, and user guidance with version control and A/B testing.
- **Semantic Search Engine:** High-relevance search system that indexes and retrieves research papers, code repositories, and datasets based on user queries with $\geq 85\%$ precision.
- **AI Validation Framework:** Automated testing suite that validates AI outputs for syntax errors, security vulnerabilities, logical consistency, and user intent alignment.
- **Model Performance Dashboard:** Real-time monitoring system tracking AI latency, token usage, cost per request, success rates, and failure modes.
- **Cost Optimization Report:** Monthly analysis showing AI cost per user, efficiency improvements, and recommendations for scaling sustainably.
- **AI Documentation:** Technical documentation covering prompt engineering best practices, model selection guidelines, and troubleshooting workflows for the team.

KPIS / EVALUATION CRITERIA

- **AI Output Quality:** $\geq 90\%$ of AI-generated projects are buildable without manual intervention
- **Response Latency:** ≤ 1.5 seconds average AI response time for 95th percentile requests
- **System Reliability:** ≤ 1 AI-related failure (timeout, error, hallucination) per week
- **Cost Efficiency:** Month-over-month reduction in AI cost per user while maintaining quality
- **User Satisfaction:** $\geq 8/10$ user rating on AI-generated recommendations and outputs

- **Search Precision:** $\geq 85\%$ precision on semantic search results (measured by user click-through and selection)
- **Model Performance:** $\geq 95\%$ uptime for AI services with automated failover

ELIGIBILITY & IDEAL CANDIDATE PROFILE

- **Educational Background:** Pursuing or completed degree in Computer Science (AI/ML specialization), Data Science, Software Engineering, Mathematics, or related fields—or demonstrated expertise in AI systems.
- **AI/ML Fundamentals:** Strong understanding of machine learning concepts, neural networks, transformers, attention mechanisms, and LLM architectures (GPT, BERT, T5, etc.).
- **Prompt Engineering:** Experience designing effective prompts, using few-shot learning, chain-of-thought reasoning, and handling edge cases in LLM outputs.
- **Programming Skills:** Proficiency in Python and AI/ML libraries (LangChain, LlamaIndex, OpenAI SDK, HuggingFace Transformers, PyTorch/TensorFlow).
- **API & Backend Knowledge:** Experience integrating LLM APIs (OpenAI, Anthropic, Google AI), handling rate limits, implementing caching, and optimizing API calls.
- **Systems Thinking:** Ability to design end-to-end AI pipelines including preprocessing, model inference, post-processing, validation, and error handling.
- **Semantic Search & RAG:** Familiarity with vector databases (Pinecone, Weaviate, ChromaDB), embeddings, retrieval-augmented generation, and knowledge graph concepts.
- **Debugging & Testing:** Strong problem-solving skills for debugging AI failures, analyzing edge cases, and implementing robust testing frameworks.
- **Tool Proficiency (Preferred):** Experience with LangChain, LangSmith, vector databases, Jupyter notebooks, Git, Docker, and cloud platforms (AWS/GCP/Azure).
- **Self-Starter Attitude:** Research-oriented mindset with ability to experiment, iterate rapidly, and implement cutting-edge AI techniques independently.
- **Bonus Points:** Published AI/ML research, contributions to open-source AI projects, Kaggle competitions, or personal projects using LLMs/agent systems.

WHY JOIN OPEN IDEA?

- **Be a Category Creator:** Help build India's first AI-powered unified innovation platform from the ground up—be part of a high-growth startup poised to redefine how innovation happens.
- **Own Your Impact:** Direct influence on the AI infrastructure that powers the entire platform—your work enables thousands of users to build without code.
- **Learn & Grow:** Exposure to production AI systems, LLM optimization, semantic search, prompt engineering at scale, and cutting-edge AI research applications.
- **Work with a Mission:** Democratize innovation and make global knowledge accessible to everyone—from students in classrooms to researchers in labs.
- **Flexible & Collaborative:** Join a lean, high-energy team where your ideas matter and execution speed is celebrated.

HOW TO APPLY

Send your resume, GitHub profile, and a brief note explaining why you're excited about Open Idea to: **info@openidea.world**

Visit our platform: **openidea.world**